COURSE 2 · SECTION 12 · WORKED SOLUTION

Rework, late gates and the valve program

Section 12 · Design program status dashboard (portfolio project 1)

DESIGN QUESTION

How many gates needed rework or finished late, where does the motorised control valve's program slip, and how did the change backlog move between the end of 2025 and mid-2026?

Steps

1. Open Engxtech_S12.pbix > File > Save as > Engxtech_S12_Exercise.
2. Right-click _Measures > New measure: Rework Gates and Revs With Rework (below). Cards for both.
3. Table view > NPD_Stages > New column: Gate Days Late (below); Whole number. Measure Late Gates. Table visual: PartRev, Stage, Gate Days Late (Don't summarize), sorted descending.
4. Product slicer = Motorised control valve: read the On-time gates card, the slip chart and the Open ECRs card. Clear.
5. Hover the backlog line at Dec 2025 and Jun 2026.
6. Part slicer = P300: add a card with COUNTROWS of closed changes, or read the table below; Median and p90 cards.

DAX new in this exercise

```
Rework Gates = CALCULATE ( COUNTROWS ( NPD_Stages ), NPD_Stages[GateResult] = "Rework" )
Revs With Rework = CALCULATE ( DISTINCTCOUNT ( NPD_Stages[PartRev] ), NPD_Stages[GateResult] = "Rework" )
NPD_Stages[Gate Days Late] = DATEDIFF ( NPD_Stages[PlannedEnd], NPD_Stages[ActualEnd], DAY )      -- column
Late Gates = CALCULATE ( COUNTROWS ( NPD_Stages ), NPD_Stages[Gate Days Late] > 0 )
Closed Changes = CALCULATE ( COUNTROWS ( Changes ), Changes[Status] = "Closed" )
```

Rework appears on Checking and Design review together, because each review loop repeats both gates: 16 rework gates on 8 revisions. 84 gates finished late, but only the stages that ran over their own plan add slip days; the others inherited an earlier delay.

The motorised control valve uses five parts (P200, P300, E600, E700, E800). Its gates are on time more often than the average, and Checking is still where it slips. The backlog rose from 15 open changes at the end of 2025 to 26 at the end of June 2026.

A sentence for the review: 'Checking is the stage that slips, also on the valve program; the open change backlog grew from 15 to 26 in the first half of 2026.'

Expected result

Check	Result
Rework gates / revisions with rework	**16 / 8**
Late gates	**84**
Latest gate	**P300-B / Production handover / 22 d**
Motorised control valve: On-time Gate %	**62.1%**
Motorised control valve: top slip stage	**Checking, 31 d**
Motorised control valve: open ECRs	**15**
Open Backlog, Dec 2025 / Jun 2026	**15 / 26**
P300: closed / median / p90 cycle days	**26 / 47.5 / 62.5**